The 3x+1 Problem For Rational Numbers

Invariance of Periodic Sequences in 3x+1 Problem

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Abstract—In the paper, we discuss a generalization of 3x+1 Problem sometimes called Collatz' (Syracuse) Conjecture. For a given initial rational number, each next number is obtained by dividing the previous integer by 2 (T operation), or multiplying it by 3, adding 1 and then dividing by 2 (S operation), or multiplying by 3, adding 2 and then dividing by 2 (V operation), or finally, just multiplying by 3, and then dividing by 2 (W operation). The presence of the last operation W makes the problem different from what was studied in the previous papers of the author. If a sequence consisted of *T*, *S*, *V* and *W* operations is given then one can ask whether there is an initial rational number x_0 for which the application of these operations in the given order will return at the end of this procedure the starting number x_0. We will also study how this rational number changes when the order of operations changes. We proved that there is an invariant which is not dependent on the order of operations. In contrast to previous papers, we developed some terminology and notations, and now the results are stated and proved in a reader friendly way. The proofs are also considerably simplified.

Index Terms—3*x*+1 problem, Collatz Conjecture, *p*-adic numbers, 3-adic numbers, periods, rational numbers, fractions, algorithm.

I. Introduction

Nicknamed as The Ultimate Challenge [1], The 3x+1 Problem is one of the longstanding problems of Computer Science and Mathematics. In essence, it says that whatever initial positive integer x_0 you choose, the recursive sequence $x_{n+1} = 3x_n + 1$ if x_n is an odd number and $x_{n+1} = \frac{x_n}{2}$ if x_n is an even number, will return $x_N = 1$ for some positive integer N. Even in its simplified form, which says that $1, 4, 2, 1, 4, 2, \dots$ is the only periodic sequence that this recursive formula can produce, the problem is still unsolved. This easily stated problem shows how complex the behavior of a simple algorithm can be. In the current paper a new approach for the last problem will be discussed [2-5].

II. Terminology and Notation

Consider the set of operations $F = \left\{T(x) = \frac{x}{2};\ S(x) = \frac{3x+1}{2};\ V(x) = \frac{3x+2}{2};\ W(x) = \frac{3x}{2}\right\}$. Suppose that a finite sequence of operations $P = B_0 B_1 \dots B_{n+m-1}$ is given, where $B_i \in F$ for $i = 0,1,2,\dots,n+m-1$ and repetitions are allowed. For the sake of simplicity, we assume that $B_{n+m} = B_0$ and in general $B_i = B_j$ if $i \equiv j \pmod{n+m}$. So, P can be extended to be a sequence infinite both to the right and to the left. Suppose also that n is the number of T operations in finite version of P and m is the number of non-Toperations in finite version of P, i.e. the number of S, V and W operations together. We assume that m is positive and n is nonnegative. Consider the equation $B_0 B_1 \dots B_{n+m-2} B_{n+m-1}(x) = x$. It is a linear equation of x and its solution x_0 is a rational number. Similarly, denote the solution of $B_1 \dots B_{n+m-2} B_{n+m-1} B_0(x) = x$ by x_1. Note that $x_0 = B_0(x_1)$. Next, denote the solution of $B_2 \dots B_{n+m-2} B_{n+m-1} B_0 B_1(x) = x$ by x_2. Note again that $x_1 = B_1(x_2)$. By continuing in this manner we determine the other numbers x_i, until we reach x_{n+m-1}, which is the solution of $B_{n+m-1} B_0 B_1 \dots B_{n+m-2}(x) = x$, for which we also have $x_{n+m-1} = B_{n+m-1}(x_0)$. For completeness, we assume that $x_{n+m} = x_0$ and in general $x_i = x_j$ if $i \equiv j \pmod{n+m}$. We will prove that these solutions x_i, where $i = 0, 1, \dots, n+m$, have some interesting properties. For this we need to introduce the numbers $U_i = \frac{2^i}{2^{n+m} - 3^m}$, where $i = 0, 1, 2, \dots, n+m$. The first number $U_0 = \frac{1}{2^{n+m} - 3^m}$ can be interpreted as a solution of $U_0 = \frac{3^m U_0 + 1}{2^{n+m}}$. There is a pair of nonnegative integers (k, j), for which $3^k U_0 \pm U_j$ is an integer, or equivalently, $3^k U_i \pm U_{i+j}$ is an integer $(0 \le i \le i+j \le n+m)$. For example, one can take trivial pairs $(k, j) = (0, 0)$ or other trivial pairs like $(k,\ j) = (m,\ n+m)$ for the – (minus) case. But usually, we also have nontrivial pairs and if there is one then there are infinitely many of them. We are especially interested with the pairs (k, j) for which $k > 0$ and k is minimal, in the sense that there is no pair (k, j) with a smaller $k > 0$. We will call such pairs *primitive* and it is possible to prove that other pairs can be generated using these pairs. Note that for this and for the other pairs (k, j) satisfying the above condition we can also say that $2^{n+m} - 3^m$ divides $3^k \pm 2^j$. Note that + (plus) and – (minus) cases are considered separately. In the sense that the pair of integers (k, j) for the + case is different from that for the – case. Similarly, there is a pair of non-negative integers (k_1, j_1), for which $3^{k_1} U_{j_1} \pm U_0$ is an integer, or equivalently, $3^{k_1} U_{i+j_1} \pm U_i$ is an integer $(0 \le i \le i + j_1 \le n+m)$. Again, one can take trivial pairs like $(k_1, j_1) = (0, 0)$ for the – (minus) case, but there are also infinitely many other pairs. Again, for the pair (k_1, j_1) we can also say that $2^{n+m} - 3^m$ divides $3^{k_1} 2^{j_1} \pm 1$. Finally, suppose that $\sigma(s,r)$, where $s \leq r$, is the number of non-T operations in the fragment $B_s B_{s+1} \dots B_{r-1}$ of the infinite sequence P. We also assume that $\sigma(s,s) = 0$.

III. Theorems and their proofs

Theorem 1. If the pair of nonnegative integers (k,j) satisfy $3^k U_0 - U_j \in Z$ then for the numbers x_i defined above, the difference $3^k x_i - 3^{\sigma(i,i+j)} x_{i+j}$ is also an integer.

Proof. The proof is based on the following simple idea. We can first show that the claim is true for one particular sequence of operations. Then we can prove that the correctness of the claim doesn't change when we change one W operation in the sequence to S, or one S operation to V. In this way we can obtain any sequence from a special sequence of operations consisted of only W and T operations. Let us start the proof. Suppose now that all S and V operations in P are replaced by W. This means that the given sequence of operations is consisted of only W and T operations. For example $P = WTWWWT$. It is obvious that in this case $x_i = 0$ for all $0 \leq i \leq n+m$. So, $3^k x_i \pm 3^{\sigma(i,i+j)} x_{i+j} = 0$ and therefore the claim is trivially true. Now, we will replace some of the W operations by S and then by V and prove that the claim remains true. Suppose that the claim is already true for a sequence of operations P. First, make a cyclic permutation of P so that the operation W which will be changed to S comes at the beginning of P. For example, suppose we are going to change the operation W with a bar above it in the sequence $P = WTW\overline{W}WT$. We do a cycling permutation to write it as $P_1 = \overline{W}WTWTW$. The numbers x_i $(0 \leq i \leq n+m)$ also make the same cyclic permutation. Let us rename them so that they are labelled again as x_i $(0 \leq i \leq n+m)$ with respect to P_1. Now consider the sequence of operations P_2 which is obtained from P_1 by replacing the W operation in the beginning by S. In the example, it will look like $P_2 = SWTWTW$. Let us denote by x_i' $(0 \leq i \leq n+m)$ the rational numbers corresponding to the sequence of operations P_2.

Let us denote by $R_i = B_{n+m-i} \dots B_{n+m-2} B_{n+m-1}$ the sequence formed by the last i operations in P_1. Denote by $R_i[b]$ the number obtained from application of the operations of the sequence R_i to some number b in the given order (from right to left). In particular, $R_0[b] = b$. As we said before, the sequence P_1 starts with W and the sequence P_2 starts with S, all the other operations are the same, that is $P_1 = WR_{n+m-1}$ and $P_2 = SR_{n+m-1}$. Since $P_1[x_0] = x_0$ and $P_2[x_0'] = x_0'$, we can write

$$x_0 = \frac{3R_{n+m-1}[x_0]}{2}, x_0' = \frac{3R_{n+m-1}[x_0'] + 1}{2},$$

and find the difference

$$x_0' - x_0 = \frac{3R_{n+m-1}[x_0' - x_0] + 1}{2} = \frac{3^m(x_0' - x_0)}{2^{n+m}} + \frac{1}{2}.$$

By solving this equation for $x_0' - x_0$, we obtain

$$x_0' = x_0 + U_{n+m-1}.$$

Using this we prove step by step that

$$x_{n+m-1}' = B_{n+m-1}(x_{n+m}') = B_{n+m-1}(x_0') =$$

$$= B_{n+m-1}(x_0 + U_{n+m-1}) =$$

$$= x_{n+m-1} + 3^{\sigma(n+m-1,n+m)} \cdot U_{n+m-2}.$$

Similarly,

$$\begin{aligned} x_{n+m-2}' = B_{n+m-2}(x_{n+m-1}') &= \\ &= B_{n+m-2}\big(x_{n+m-1} + 3^{\sigma(n+m-1,n+m)} \\ &\cdot U_{n+m-2}\big) = \end{aligned}$$

$$= x_{n+m-2} + 3^{\sigma(n+m-2,n+m)} \cdot U_{n+m-3}.$$

By continuing in this manner we obtain that

$$x_{n+m-k}' = x_{n+m-k} + 3^{\sigma(n+m-k,n+m)} \cdot U_{n+m-k-1},$$

for all $k = 0,1,2,\dots,n+m-1$. In particular, for $k = n+m-1$ we obtain $x_1' = x_1 + 3^{\sigma(1,n+m)} \cdot U_0$. Noting the fact that $\sigma(1,n+m) = m-1$, we can write $x_1' = x_1 + 3^{m-1} \cdot U_0$. It is also possible to write the above equalities with simpler indices: $x_i' = x_i + 3^{\sigma(i,n+m)} \cdot U_{i-1}$, where $i = 1,2,\dots,n+m$. Going one more step and using the fact that B_0 is not a T operation, we obtain again

$$\begin{aligned} x_0' = B_0(x_1') = B_0(x_1 + 3^{m-1} \cdot U_0) = x_0 + \frac{1}{2}(3^m \cdot U_0 + 1) \\ = x_0 + U_{n+m-1}. \end{aligned}$$

This can also be interpreted as

$$x_{n+m}' = x_{n+m} + U_{n+m-1}.$$

Suppose now that the pair of nonnegative integers (k,j) satisfy $3^k U_0 - U_j \in Z$ and for the numbers x_i $(0 \leq i \leq n+m)$ the difference $3^k x_i - 3^{\sigma(i,i+j)} x_{i+j}$ is an integer. We want to prove that the last property holds true for the numbers x_i' $(0 \leq i \leq n+m)$, too. Assume first that $i > 0$. Using the above results, we can write

$$\begin{aligned} 3^k x_i' - 3^{\sigma(i,i+j)} x_{i+j}' & \\ &= 3^k\big(x_i + 3^{\sigma(i,n+m)} \cdot U_{i-1}\big) - 3^{\sigma(i,i+j)} \\ &\cdot \big(x_{i+j} + 3^{\sigma(i+j,n+m)} \cdot U_{i+j-1}\big) \\ &= 3^k x_i - 3^{\sigma(i,i+j)} x_{i+j} + 3^{\sigma(i,n+m)} \\ &\cdot \big(3^k \cdot U_{i-1} - U_{i+j-1}\big), \end{aligned}$$

which is an integer by assumption. For the case $i = 0$ we obtain

$$\begin{aligned} 3^k x_0' - 3^{\sigma(0,j)} x_j' = 3^k(x_0 + U_{n+m-1}) - 3^{\sigma(0,j)} \\ \cdot \big(x_j + 3^{\sigma(j,n+m)} \cdot U_{j-1}\big) = \end{aligned}$$

$$= \left(3^k x_0 - 3^{\sigma(0,j)} x_j\right) + \left(3^k \cdot U_{n+m-1} - 3^{\sigma(0,n+m)} \cdot U_{j-1}\right),$$

where the summand $3^k x_0 - 3^{\sigma(0,j)} x_j$ is again an integer by assumption. For the second summand we can write

$$3^k \cdot U_{n+m-1} - 3^{\sigma(0,n+m)} \cdot U_{j-1} = 3^k \cdot U_{n+m-1} - 3^m \cdot U_{j-1}$$
$$= \frac{3^k}{2}\left(3^m \cdot U_0 + 1\right) - 3^m \cdot U_{j-1}$$
$$= \frac{3^m\left(3^k \cdot U_0 - U_j\right) + 3^k}{2},$$

which is again an integer, because by assumption $3^k \cdot U_0 - U_j \in Z$ and therefore $3^m\left(3^k \cdot U_0 - U_j\right) + 3^k$ is even.

The case when S operation is replaced by V is considered in a similar way.

Proof is complete.

The following theorems are proved analogously.

Theorem 2. If the pair of nonnegative integers (k, j) satisfy $3^k U_0 + U_j \in Z$ then for the numbers x_i defined above, the sum $3^k x_i + 3^{\sigma(i,i+j)} x_{i+j}$ is also an integer.

Theorem 3. If the pair of nonnegative integers (k_1, j_1) satisfy $3^{k_1} U_{j_1} - U_0 \in Z$ then for the numbers x_i defined above, the difference $3^{k_1+\sigma(i,i+j_1)} x_{i+j_1} - x_i$ is also an integer.

Theorem 4. If the pair of nonnegative integers (k_1, j_1) satisfy $3^{k_1} U_{j_1} + U_0 \in Z$ then for the numbers x_i defined above, the sum $3^{k_1+\sigma(i,i+j_1)} x_{i+j_1} + x_i$ is also an integer.

IV. Examples

Consider the sequence of operations $P = B_0 B_1 B_2 B_3 B_4 B_5 = TTSWVW$. Here $m = 4$ and $n = 2$. This sequence can be obtained from starting sequence of operations $TTWWWW$ in the following way. The bar indicates the operation that is changed in the next sequence.

$$TT\bar{W}WWW \to TTSW\bar{W}W \to TTSW\bar{S}W \to TTSWVW$$

The solution of the linear equation $TTSWVW(x) = x$ is the number $x_6 = x_0 = -44/17$. We can also find the other numbers $x_1 = -88/17, x_2 = -176/17, x_3 = -123/17, x_4 = -82/17, x_5 = -66/17$. We also observe that for the numbers $U_i = -2^i/17$ $(i = 0,1,\dots,5)$ the following holds true:

a) $3^6 \cdot U_0 - U_5 = -41 \in Z$, ($k = 6, j = 5$ in Theorem 1)

b) $3^2 \cdot U_0 + U_3 = -1 \in Z$, ($k = 2, j = 3$ in Theorem 2)

c) $3^2 \cdot U_1 - U_0 = -1 \in Z$, ($k_1 = 2, j_1 = 1$ in Theorem 3)

d) $3^2 \cdot U_5 + U_0 = -17 \in Z$, ($k_1 = 2, j_1 = 5$ in Theorem 4)

Now we will experiment with numbers x_i $(i = 0,1,\dots,5)$.

a) $3^6 \cdot x_0 - 3^{\sigma(0,5)} \cdot x_5 = 3^6 \cdot \left(-\frac{44}{17}\right) - 3^3 \cdot \left(-\frac{66}{17}\right) = -1782 \in Z$. Note that $\sigma(0,5) = 3$ is the number of non-T operations in the fragment $B_0 B_1 B_2 B_3 B_4 = TTSWV$. (cf. Theorem 1)

b) $3^2 \cdot x_2 + 3^{\sigma(2,5)} \cdot x_5 = 3^2 \cdot \left(-\frac{176}{17}\right) + 3^3 \cdot \left(-\frac{66}{17}\right) = -198 \in Z$. (cf. Theorem 2)

c) $3^{2+\sigma(3,4)} \cdot x_4 - x_3 = 3^3 \cdot \left(-\frac{82}{17}\right) - \left(-\frac{123}{17}\right) = -123 \in Z$. (cf. Theorem 3)

d) $3^{2+\sigma(1,6)} \cdot x_6 + x_1 = 3^6 \cdot \left(-\frac{44}{17}\right) + \left(-\frac{88}{17}\right) = -1892 \in Z$. (cf. Theorem 4)

V. Program in Pascal

The following program [6] written in Pascal language can help to explore more patterns and find more examples. By changing the string P:='WVWSTT' before the compilation one can study any sequence P. Before using the program, note that in the program the sequence of operations is written in the reverse order.

```
program threexpluszerooneortwo;
uses crt, sysutils;
var k, i, m, n, u: integer;
var P, w: string;
begin
  P:='WVWSTT'; k:=Length(P); i:=0; m:=1; n:=1; u:=0;
while i<k do
begin
   i:=i+1; w:=RightStr(P,1); delete(P,length(P),1);
   if w = 'S' then
    begin
     m:=2*m; u:=2*u-n; n:=n*3;
    end
   else
    begin
     if w = 'V' then
      begin
       m:=2*m; u:=2*u-2*n; n:=n*3;
      end
      else
     begin
      if w = 'W' then
       begin
        m:=2*m; u:=2*u; n:=n*3;
       end
      else
       begin
        m:=2*m; u:=2*u;
       end
     end
    end
end;
writeln('x=', u, '/', n-m);
```

end.

VI. *P*-ADIC REPRESENTATIONS

It is also possible to interpret these results in the context of p-adic numbers. If the numbers x_i $(0 \le i \le n + m)$ are written as 3-adic numbers in a special table then the above results will have a nice visualization. Let us return to the example $P = TTSWVW$.

a) The underlined digits explain the observation that $3^6 \cdot x_0 - 3^3 \cdot x_5 \in Z$, which was mentioned above. The digits in the underlined part are repeated and this is true for any similarly aligned parts of the matrix.

$$\begin{array}{lll}
x_6 = & \ldots\ 1\ \underline{2\ 2}\ 1\ 1\ 0\ 2\ 0\ 0\ 2\ 2\ 2\ |0 & W = B_5 \\
x_5 = & \ldots\ 2\ 1\ 1\ 0\ 2\ 0\ 1\ 0\ 0\ 1\ 1\ 1\ 0\ |2 & V = B_4 \\
x_4 = & \ldots\ 1\ 0\ 2\ 0\ 1\ 0\ 0\ 1\ 1\ 2\ 0\ 2\ 0\ 1\ |0 & W = B_3 \\
x_3 = & \ldots\ 2\ 0\ 1\ 0\ 0\ 1\ 1\ 2\ 0\ 2\ 1\ 2\ 1\ 2\ 0\ |1 & S = B_2 \\
x_2 = & \ldots\ 1\ 0\ 0\ 1\ 1\ 2\ 0\ 2\ 1\ 2\ 2\ 1\ 0\ 2\ 1\ 2 & T = B_1 \\
x_1 = & \ldots\ 0\ 1\ 1\ 2\ 0\ 2\ 1\ \underline{2\ 2}\ 1\ 1\ 0\ 1\ 2\ 2\ 1 & T = B_0 \\
x_0 = & \ldots\ 1\ 2\ 0\ 2\ 1\ 2\ 2\ 1\ 1\ 0\ 2\ 0\ 0\ 2\ 2\ 2 &
\end{array}$$

One can also find interpretation with digits for the other observations above.

b) $3^2 \cdot x_2 + 3^3 \cdot x_5 \in Z$. The digits in the underlined parts are inverted, in the sense that 0's are replaced by 2's and vice versa, while 1's remain unchanged.

$$\begin{array}{lll}
x_6 = & \ldots\ 1\ 2\ 2\ 1\ 1\ 0\ 2\ 0\ 0\ 2\ 2\ 2\ |0 & W = B_5 \\
x_5 = & \ldots\ 2\ \underline{1\ 1}\ 0\ 2\ 0\ 1\ 0\ 0\ 1\ 1\ 1\ 0\ |2 & V = B_4 \\
x_4 = & \ldots\ 1\ 0\ 2\ 0\ 1\ 0\ 0\ 1\ 1\ 2\ 0\ 2\ 0\ 1\ |0 & W = B_3 \\
x_3 = & \ldots\ 2\ 0\ 1\ 0\ 0\ 1\ 1\ 2\ 0\ 2\ 1\ 2\ 1\ 2\ 0\ |1 & S = B_2 \\
x_2 = & \ldots\ 1\ 0\ 0\ \underline{1\ 1}\ 2\ 0\ 2\ 1\ 2\ 2\ 1\ 0\ 2\ 1\ 2 & T = B_1 \\
x_1 = & \ldots\ 0\ 1\ 1\ 2\ 0\ 2\ 1\ 2\ 2\ 1\ 1\ 0\ 1\ 2\ 2\ 1 & T = B_0 \\
x_0 = & \ldots\ 1\ 2\ 0\ 2\ 1\ 2\ 2\ 1\ 1\ 0\ 2\ 0\ 0\ 2\ 2\ 2 &
\end{array}$$

c) $3^3 \cdot x_4 - x_3 \in Z$

$$\begin{array}{lll}
x_6 = & \ldots\ 1\ 2\ 2\ 1\ 1\ 0\ 2\ 0\ 0\ 2\ 2\ 2\ |0 & W = B_5 \\
x_5 = & \ldots\ 2\ 1\ 1\ 0\ 2\ 0\ 1\ 0\ 0\ 1\ 1\ 1\ 0\ |2 & V = B_4 \\
x_4 = & \ldots\ 1\ 0\ 2\ 0\ 1\ \underline{0\ 0}\ 1\ 1\ 2\ 0\ 2\ 0\ 1\ |0 & W = B_3 \\
x_3 = & \ldots\ 2\ 0\ 1\ \underline{0\ 0}\ 1\ 1\ 2\ 0\ 2\ 1\ 2\ 1\ 2\ 0\ |1 & S = B_2 \\
x_2 = & \ldots\ 1\ 0\ 0\ 1\ 1\ 2\ 0\ 2\ 1\ 2\ 2\ 1\ 0\ 2\ 1\ 2 & T = B_1 \\
x_1 = & \ldots\ 0\ 1\ 1\ 2\ 0\ 2\ 1\ 2\ 2\ 1\ 1\ 0\ 1\ 2\ 2\ 1 & T = B_0 \\
x_0 = & \ldots\ 1\ 2\ 0\ 2\ 1\ 2\ 2\ 1\ 1\ 0\ 2\ 0\ 0\ 2\ 2\ 2 &
\end{array}$$

d) $3^6 \cdot x_6 + x_1 \in Z$.

$$\begin{array}{lll}
x_6 = & \ldots\ 1\ 2\ 2\ 1\ \underline{1\ 0}\ 2\ 0\ 0\ 2\ 2\ 2\ |0 & W = B_5 \\
x_5 = & \ldots\ 2\ 1\ 1\ 0\ 2\ 0\ 1\ 0\ 0\ 1\ 1\ 1\ 0\ |2 & V = B_4 \\
x_4 = & \ldots\ 1\ 0\ 2\ 0\ 1\ 0\ 0\ 1\ 1\ 2\ 0\ 2\ 0\ 1\ |0 & W = B_3 \\
x_3 = & \ldots\ 2\ 0\ 1\ 0\ 0\ 1\ 1\ 2\ 0\ 2\ 1\ 2\ 1\ 2\ 0\ |1 & S = B_2 \\
x_2 = & \ldots\ 1\ 0\ 0\ 1\ 1\ 2\ 0\ 2\ 1\ 2\ 2\ 1\ 0\ 2\ 1\ 2 & T = B_1 \\
x_1 = & \ldots\ 0\ 1\ \underline{1\ 2}\ 0\ 2\ 1\ 2\ 2\ 1\ 1\ 0\ 1\ 2\ 2\ 1 & T = B_0 \\
x_0 = & \ldots\ 1\ 2\ 0\ 2\ 1\ 2\ 2\ 1\ 1\ 0\ 2\ 0\ 0\ 2\ 2\ 2 &
\end{array}$$

For more details about the 3-adic representations of these numbers and how they are constructed, we recommend the previous papers [2-5].

VII. CONCLUSION

These results show that there are invariants, determined by the numbers U_i, which are independent of the order and nature of operations in P. The proof that we used is based on the finite variant of the Method of Mathematical Induction. It should be mentioned that the main motivation behind this study was the determination of all possible sequences P of operations T, S, W, V for which $x_0 \in Z$. This happens when for example $P = WVWSTT$.

$$\begin{array}{llll}
x_6 = & -12 = & \cdots\ 2\ 2\ 2\ 2\ 2\ 2\ 2\ 2\ 2\ 1\ 2\ 0 & T = B_5 \\
x_5 = & -6 = & \cdots\ 2\ 2\ 2\ 2\ 2\ 2\ 2\ 2\ 2\ 2\ 1\ 0 & T = B_4 \\
x_4 = & -3 = & \cdots\ 2\ 2\ 2\ 2\ 2\ 2\ 2\ 2\ 2\ 2\ 2\ 0\ |1 & S = B_3 \\
x_3 = & -4 = & \cdots\ 2\ 2\ 2\ 2\ 2\ 2\ 2\ 2\ 2\ 2\ 2\ 1\ 2\ |0 & W = B_2 \\
x_2 = & -6 = & \cdots\ 2\ 2\ 2\ 2\ 2\ 2\ 2\ 2\ 2\ 2\ 2\ 2\ 1\ 0\ |2 & V = B_1 \\
x_1 = & -8 = & \cdots\ 2\ 2\ 2\ 2\ 2\ 2\ 2\ 2\ 2\ 2\ 2\ 2\ 2\ 0\ 1\ |0 & W = B_0 \\
x_0 = & -12 = & \cdots\ 2\ 2\ 2\ 2\ 2\ 2\ 2\ 2\ 2\ 2\ 2\ 2\ 2\ 1\ 2\ 0 &
\end{array}$$

Here the numbers on the left are written in decimal number system. The determination of all possible integer cases such as this, within all possible sequences consisted of T, S, V and W operations is an open question. We also don't know whether the number of such integer cases is finite or not. At the end, I would like to mention inspiring paper [7] by John H. Conway (1937-2020) whose generalizations of 3x+1 problem largely inspired the previous papers by the author and the current study.

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